

# Rajalakshmi Engineering College

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## 2024\_28\_III\_OOPS Using Java Lab

### 2028\_REC\_OOPS using Java\_Week 6\_Q1

Attempt : 1  
Total Mark : 10  
Marks Obtained : 10

#### Section 1 : Coding

##### 1. Problem Statement

Elsa subscribes to a premium service with a base monthly cost, a service tax and an extra feature cost. Assist her in writing an inheritance program that takes input for these values and calculates the total monthly cost.

Refer to the below class diagram:

##### ***Input Format***

The first line of input consists of a double value, representing the base monthly cost.

The second line consists of a double value, representing the service tax.

The third line consists of a double value, representing the extra feature cost.

### ***Output Format***

The output prints "Rs. X" where X is a double value, rounded off to two decimal places.

Refer to the sample output for formatting specifications.

### ***Sample Test Case***

Input: 10.0

2.5

5.0

Output: Rs. 17.50

### ***Answer***

```
import java.util.Scanner;
```

```
// You are using Java
```

```
/**
```

```
 * Base class representing a subscription with a base cost.
```

```
 */
```

```
class Subscription {
```

```
    // The base cost of the subscription, accessible by subclasses.
```

```
    protected double baseMonthlyCost;
```

```
    /**
```

```
     * Constructor to initialize the Subscription object.
```

```
     * @param baseMonthlyCost The base monthly cost.
```

```
     */
```

```
    public Subscription(double baseMonthlyCost) {
```

```
        this.baseMonthlyCost = baseMonthlyCost;
```

```
    }
```

```
}
```

```
/**
```

```
 * Represents a subscription that includes a service tax, inheriting from  
Subscription.
```

```
 */
```

```
class TaxedSubscription extends Subscription {
```

```
    // The service tax amount, accessible by subclasses.
```

```

protected double serviceTax;

/**
 * Constructor to initialize the TaxedSubscription object.
 * It calls the parent class constructor to set the base cost.
 * @param baseMonthlyCost The base monthly cost.
 * @param serviceTax The service tax amount.
 */
public TaxedSubscription(double baseMonthlyCost, double serviceTax) {
    super(baseMonthlyCost); // Call the constructor of the parent class
    (Subscription)
    this.serviceTax = serviceTax;
}
}

/**
 * Represents a premium subscription with extra features, inheriting from
 * TaxedSubscription.
 */
class PremiumSubscription extends TaxedSubscription {
    // The cost for extra features.
    protected double extraFeatureCost;

    /**
     * Constructor to initialize the PremiumSubscription object.
     * It calls the parent class constructor to set the base cost and service tax.
     * @param baseMonthlyCost The base monthly cost.
     * @param serviceTax The service tax amount.
     * @param extraFeatureCost The cost of extra features.
     */
    public PremiumSubscription(double baseMonthlyCost, double serviceTax,
    double extraFeatureCost) {
        super(baseMonthlyCost, serviceTax); // Call the constructor of the parent
    class (TaxedSubscription)
        this.extraFeatureCost = extraFeatureCost;
    }

    /**
     * Calculates the total monthly cost by summing the base cost, service tax,
     * and extra feature cost, all of which are available in this class through
     * inheritance.
     * @return The total monthly subscription cost.

```

```

    */
    public double calculateMonthlyCost() {
        return baseMonthlyCost + serviceTax + extraFeatureCost;
    }
}

public class Main {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        double baseMonthlyCost = scanner.nextDouble();
        double serviceTax = scanner.nextDouble();
        double extraFeatureCost = scanner.nextDouble();

        PremiumSubscription premiumSubscription = new
PremiumSubscription(baseMonthlyCost, serviceTax, extraFeatureCost);

        double totalMonthlyCost = premiumSubscription.calculateMonthlyCost();

        System.out.printf("Rs. %.2f%n", totalMonthlyCost);

        scanner.close();
    }
}

```

**Status :** Correct

**Marks :** 10/10